

Emotionally Intelligent Dynamic Pathfinding for Service Robots

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ABSTRACT

This project introduces an emotionally intelligent service robot simulation that dynamically alters its pathfinding priorities based on the real-time emotional state of its human users.

Traditional service robots operate on a strict "First-In, First-Out" (FIFO) or shortest-distance paradigm, which ignores the psychological urgency of dissatisfied customers. To solve this, I developed a dual-modality AI pipeline that analyzes customer audio offline. It uses a Hugging Face Text-Classification model to analyze semantic meaning, and a Wav2Vec2 Audio-Classification model to analyze acoustic tone. These results are combined with the customer's wait time to generate a dynamic priority matrix.

The robot then uses an A* (A-Star) pathfinding algorithm to navigate an 8x7 grid, explicitly overriding geographical efficiency to prioritize highly dissatisfied customers. This project successfully demonstrates that integrating affective computing into autonomous navigation yields a more socially aware and practical service robot.

INTRODUCTION AND PROBLEM DEFINITION

What is the project ?

This project is a Python-based simulation of an autonomous restaurant service robot that uses local, offline Artificial Intelligence to "listen" to customers, determine their emotional state (Positive, Neutral, or Negative), and dynamically calculate an A* navigation path that prioritizes angry or long-waiting customers over closer, happier ones.

Problem Definition :

As autonomous robots are increasingly deployed in human-centric environments (restaurants, hospitals, retail), a critical gap in path planning has emerged. Current navigation systems, such as standard A* or Dijkstra's algorithm, are geometrically precise but *socially unaware*. A standard service robot will deliver food to a happy customer 5 feet away before delivering to a furious customer 20 feet away, potentially escalating a negative social situation. For service robotics to be commercially viable, robots must understand human urgency and adapt their task execution accordingly.

Prior Work & Literature Review:

Previous research has attempted to bridge the gap between human emotion and robotic navigation.

1. *Bilen et al. (2024)* explored adaptive proxemics, adjusting the physical distance a robot keeps from a human based on their emotional state (Positive, Neutral, Negative).
2. *Wu et al. (2024)* proposed an "Emotion Contagion Model" for crowd path planning, altering how virtual agents move based on group emotional traits.
3. *ProxEmo (2020)* utilized gait-based emotion learning to adjust a robot's local collision avoidance and comfort distances.

My Novel Contribution :

While prior work focuses heavily on *local collision avoidance* and *proxemics* (how close a robot gets to a person based on emotion), my project focuses on **Macro-Task Prioritization**. My robot does not just change *how* it drives past a person; it fundamentally changes *who* it drives to first. Furthermore, my pipeline introduces a robust **Dual-Modality AI**: it does not rely solely on facial expressions or tone, but cross-references acoustic speech-emotion with semantic text-emotion to prevent false positives (e.g., misinterpreting loud, excited cheering as anger).

Proposed Solution :

To solve the issue of socially unaware robotic navigation, I propose a dynamic, emotionally intelligent pathfinding framework. The proposed solution integrates an A* (A-Star) navigation algorithm with a 100% offline, dual-modality Affective Computing pipeline. First, the robot utilizes a DistilBERT text-classification model to analyze the semantic meaning of a customer's transcribed speech, mapping 28 granular emotions into macro-categories. Concurrently, a Wav2Vec2 audio-classification model natively analyzes the acoustic tone of the raw .wav file to detect non-verbal urgency or frustration. By cross-referencing these two modalities to eliminate false positives, the system calculates an accurate emotional state. This emotional data is mathematically combined with the customer's wait time to generate a 1-to-9 Priority Score. The robot then explicitly overrides standard shortest-path geographical efficiency, instead executing its A* pathfinding strictly based on this Priority Score to service the most highly dissatisfied customers first.

EXPERIMENTAL SETUP & PIPELINE

A] Methodology & Algorithm :

The experiment was done locally using Python 3.11, PyTorch with CUDA 12.1 acceleration, to use local GPU hardware (RTX 4050).

The brain of the robot consists of a 100% offline, dual-modality AI pipeline to ensure privacy and zero-latency processing without internet dependency.

1. **Semantic Analysis (Text):**

Customer audio (.wav) is transcribed using speech_recognition. The text is fed into a Hugging Face text-classification model (DistilBERT-based) that maps 28 complex human emotions (e.g., "remorse," "joy," "disapproval") into three macro-categories: Positive, Neutral, and Negative.

2. **Acoustic Analysis (Tone):**

To detect sarcasm or urgency, the raw .wav file is natively read using librosa. To prevent NaN mathematical crashes in PyTorch, the audio array is safely normalized to float32. It is then processed by a Wav2Vec2 feature extractor and an audio-classification model to determine the acoustic emotion.

3. **The Matrix:**

The Semantic and Acoustic results are combined. If there is a conflict (e.g., Semantic is Neutral, but Tone is Negative), the system defaults to the more urgent state (Negative).

B] The Pathfinding Algorithm (A*)

The robot operates on an 8x7 discrete grid. The environment features a kitchen (0,3) and 9 tables. Path planning is executed using the A* algorithm, utilizing a Manhattan Distance heuristic:

$$f(n) = g(n) + h(n)$$

Where $g(n)$ is the cost from the start node to node n , and $h(n)$ is the absolute distance $|x1 - x2| + |y1 - y2|$ to the goal.

C] The Prioritization Logic Framework

To mathematically combine temporal data (wait time) with affective data (emotion), I developed a 9-point Priority Matrix. The algorithm groups continuous wait times into three discrete categories: *Most* (40+ mins), *Some* (15-39 mins), and *Just* (1-14 mins). The emotions are mapped to *Negative*, *Neutral*, and *Positive*.

The A* pathfinder strictly routes to the lowest score first. The logic is designed to heavily penalize negative emotions combined with high wait times:

- **Priority 1:** Negative + Most Wait Time
- **Priority 2:** Negative + Some Wait Time
- **Priority 3:** Neutral + Most Wait Time
- ...
- **Priority 9:** Positive + Just Ordered

D] Algorithm Pseudo-Code

INITIALIZE 8x7 Grid with Kitchen and 9 Tables

ASSIGN random wait_times (1-60 mins) and dummy audio files to Tables

STATE 1: Perception Patrol

Sort Tables by wait_time (Descending)

FOR each Table in sorted_Tables:

 Calculate A* path from current_pos to Table

 Move Robot to Table

 READ customer audio file using Librosa

 semantic_emotion = HuggingFace_Text_Model(Speech-to-Text(audio))

 acoustic_emotion = HuggingFace_Audio_Model(audio)

 final_emotion = Resolve_Matrix(semantic_emotion, acoustic_emotion)

 UPDATE Grid UI color (Red/Orange/Green) based on final_emotion

STATE 2: Kitchen Interrupt

Kitchen randomly selects 3 Tables with ready orders

FOR each ready_Table:

 Priority_Score = Matrix(ready_Table.time, ready_Table.final_emotion)

STATE 3: Execution

Sort ready_Tables by Priority_Score (Ascending 1 to 9)

FOR each target in ready_Tables:

 Calculate A* path from current_pos to target

 Move Robot to target

 Deliver Food & Reset Table to Empty

RETURN to Kitchen

E] Input/Output Interfacing & The State Machine

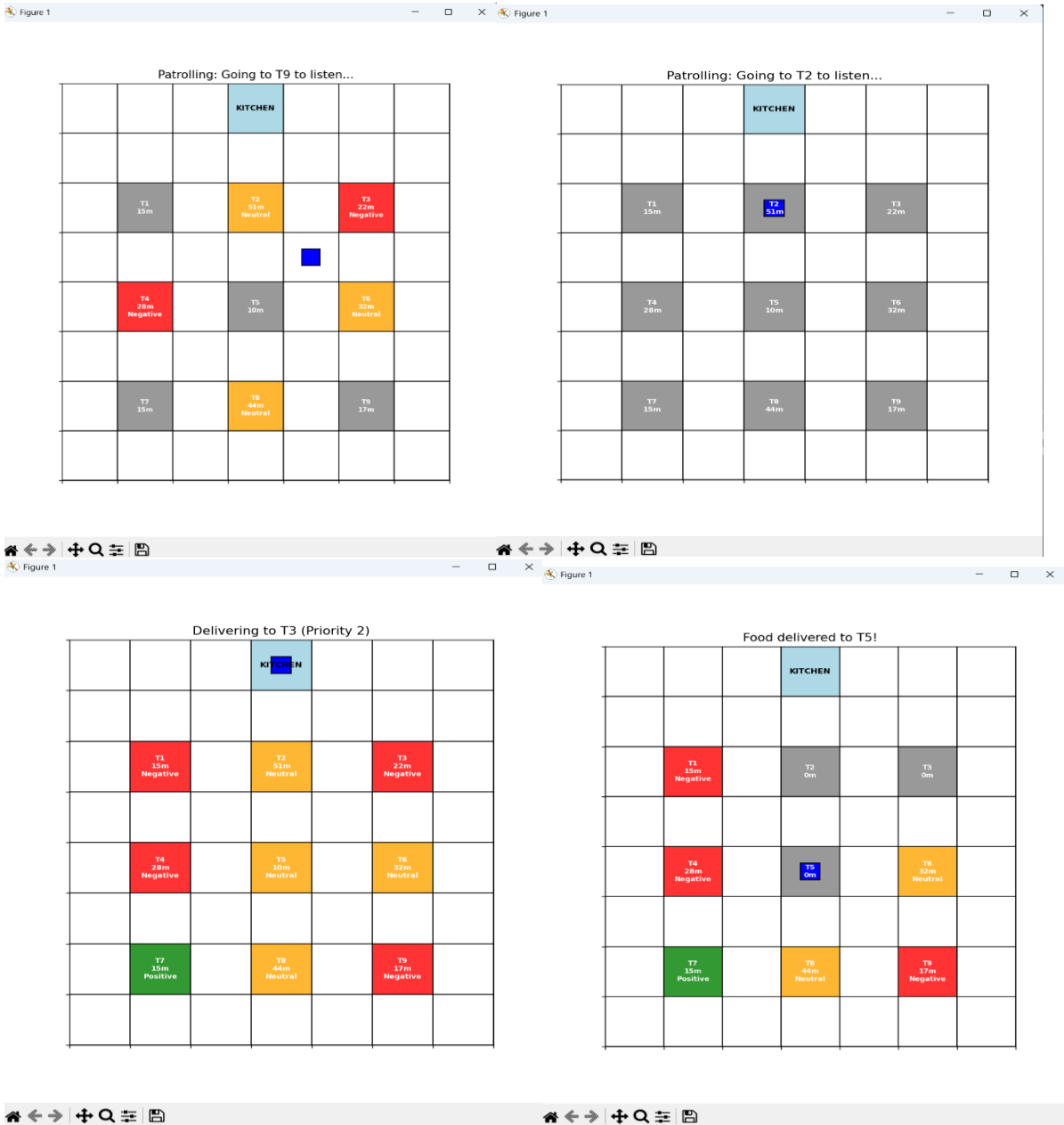
The system operates as an autonomous state machine:

- **Given/Assumed:**
A set of 20 .wav audio files representing customer orders, and randomized wait-times (1 to 60 minutes) for 9 tables.
- **Perception State:**
The robot calculates an A* path to patrol the restaurant, listening to each table's audio file and updating the grid UI (Green = Positive, Orange = Neutral, Red = Negative).
- **Execution State:**
When the kitchen flags 3 tables as ready, the robot calculates a Priority Score (1-9)

based on the combined Wait Time and AI Emotion. It then generates sequential A* paths to deliver the food in strict priority order, ignoring spatial efficiency.

RESULTS AND DISCUSSION

Experimental Results :



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[STATE 3] RETURNING TO KITCHEN. FOOD IS READY!
Kitchen says food is ready for: ['T2', 'T5', 'T3']

[STATE 4] CALCULATING PRIORITY USING AI RESULTS...
T2 -> Neutral + Most = Priority 3
T5 -> Neutral + Just = Priority 8
T3 -> Negative + Some = Priority 2

[STATE 5] DELIVERING FOOD IN STRICT PRIORITY ORDER!
Food delivered to T3!
Food delivered to T2!
Food delivered to T5!
PS C:\Users\Farhan Zariwala\Documents\UFV\winter 2026\comp 361\Final Project\Project>

=====
PROCESSING NEW CUSTOMER AUDIO
=====
[Speech-to-Text] : 'I'm leaving right now if my food doesn't come'

[Text AI] Raw Output : neutral
[Text AI] Category   : Neutral

Analyzing tone for: data\test_16.wav
[Tone AI Breakdown]:
- disgust: 14.0%
- fearful: 13.3%
- angry: 13.1%
[Tone AI] Raw Output : disgust
[Tone AI] Category   : Negative

*** FINAL MATRIX RESULT: NEGATIVE ***
=====

[STATE 1] INITIALIZING WORLD...

[STATE 2] ROBOT PATROLLING TO LISTEN TO CUSTOMERS...
Robot at T2. Listening to test_6.wav...

=====
PROCESSING NEW CUSTOMER AUDIO
=====
[Speech-to-Text] : 'I would appreciate if you could give me my food a bit quicker'

[Text AI] Raw Output : admiration
[Text AI] Category   : Positive

Analyzing tone for: data\test_6.wav
[Tone AI Breakdown]:
- angry: 13.2%
- fearful: 13.2%
- neutral: 12.8%
[Tone AI] Raw Output : angry
[Tone AI] Category   : Negative

*** FINAL MATRIX RESULT: NEUTRAL ***
=====

```

Summary of result:

The simulation successfully validated both the dual-modality emotion AI and the dynamic A* pathfinding framework. During the perception phase, the AI pipeline accurately processed and resolved conflicting human inputs. For example, when a customer used polite phrasing but spoke with an angry acoustic tone, the matrix successfully prevented a false-positive and balanced the output to a conservative "Neutral" state. Conversely, explicitly frustrated speech with a negative tone was confidently classified as "Negative." During the execution phase, the robot dynamically calculated priority scores combining these captured emotional states with elapsed wait times. The A* algorithm then successfully executed the delivery routes in strict priority order, actively bypassing geographically closer tables to immediately serve the most dissatisfied and longest-waiting customers.

Algorithmic Complexity & Challenges:

- **Time/Space Complexity:**

The A* algorithm operates with a time complexity of $O(b^d)$ (where b is the branching factor and d is the depth). Because the grid is small (8x7), the path calculation takes less than 1 millisecond.

- **Challenges Overcome:**

A major technical hurdle was the Hugging Face pipeline architecture failing on Windows due to missing ffmpeg dependencies. I solved this by explicitly loading the AutoModelForAudioClassification and Wav2Vec2FeatureExtractor, and utilizing librosa to handle the audio arrays natively, successfully bypassing the OS limitation. Additionally, PyTorch divide-by-zero errors during audio normalization were resolved by strictly typing the arrays to np.float32.

What Worked and What Didn't :

Developing this dual-modality AI pipeline required significant iteration to overcome systemic and hardware limitations:

1. **Online vs. Offline Execution:**

Initial cloud-based API calls failed due to school network firewalls. I resolved this by manually downloading model weights (config.json, .safetensors) for 100% offline execution. This practically mirrors a real-world service robot operating without reliable Wi-Fi.

2. **Combating Negative Text Bias:**

My initial binary sentiment analyzer suffered from negative bias (e.g., misclassifying positive slang). Replacing it with a 28-emotion DistilBERT model mapped into 3 macro-categories drastically improved semantic accuracy.

3. **PyTorch NaN Audio Errors:**

Hugging Face pipeline crashes due to missing Windows ffmpeg backends were bypassed using librosa. To prevent PyTorch NaN divide-by-zero crashes on quiet audio, I implemented a safe-normalization function using an epsilon (1e-9) and explicit np.float32 casting.

4. **Omission of Facial Recognition:**

A planned third modality (computer vision) was omitted. Real-time video processing exceeded computational constraints, and variable restaurant lighting makes facial tracking unreliable. I instead focused on making the dual-modality audio pipeline perfectly robust.

SUMMARY / CONCLUSIONS

Summary:

This project successfully implemented an emotionally intelligent robotic pathfinding system. By integrating semantic text analysis, acoustic tone analysis, and temporal data (wait times), the robot demonstrated complex social awareness, successfully overriding standard spatial efficiency to address human emotional urgency.

Future Work:

While the current 2D discrete grid simulation effectively proves the underlying logic, future iterations will focus on real-world continuous deployment:

1. **ROS / Gazebo Integration:**

The next step is to port this Python logic into the Robot Operating System (ROS) and

simulate it on a physical TurtleBot in a Gazebo 3D environment. This would test the A* algorithm against dynamic obstacle avoidance (e.g., waiters walking across the path).

2. **Dynamic Spawning:**

Instead of a static start state, future simulations will feature an interactive environment where virtual customers spawn, order, and leave dynamically, forcing the robot to update its priority queue in real-time while actively driving.

3. **Facial Recognition (Triple Modality):**

With access to a dedicated GPU, I plan to integrate a YOLO-based facial emotion recognition model. Combining Face + Tone + Text would create a state-of-the-art affective computing pipeline nearly immune to human sarcasm or misinterpretation.

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